GRADE 9 PRE-TECHNICAL STUDIES
CONTINUOUS ASSESSMENT TEST 2

TERM 1 | TIME: 1 HOUR 15 MINUTES | TOTAL: 50 MARKS

Name: ___________________________________ Adm. No.: ______________

School: __________________________________ Date: __________________

INSTRUCTIONS

- Answer all questions.
- Show clear steps where calculations are required.
- Use neat, labelled sketches where appropriate.
- Observe workshop safety in all practical situations.

SECTION A: SHORT-ANSWER QUESTIONS (20 MARKS)

1. Define the term specification in product design. (2 marks)

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2. State TWO reasons for selecting appropriate materials before making a product. (2 marks)

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3. Name TWO tools used for marking out materials in a workshop. (2 marks)

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4. State TWO properties of timber that make it suitable for furniture making. (2 marks)

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5. Differentiate between a first-aid box and a fire extinguisher in terms of their purpose. (2 marks)

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6. Give TWO examples of electrical energy being converted into other forms of energy in the home. (2 marks)

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7. State TWO ways of reducing risks when working with electrical equipment. (2 marks)

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8. What is meant by the term quality control? (2 marks)

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9. State TWO reasons for finishing the surfaces of a wooden product. (2 marks)

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10. Give TWO environmental considerations when disposing of workshop waste. (2 marks)

SECTION B: STRUCTURED QUESTIONS (30 MARKS)

11. Product design and material selection (8 marks)

(a) Explain THREE factors to consider when selecting materials for a product. (3 marks)

(b) Outline THREE stages involved in producing a product from a working drawing. (3 marks)

(c) State TWO ways of checking whether a finished product meets its specifications. (2 marks)

12. Workshop safety and first aid (8 marks)

(a) State FOUR common causes of accidents in a workshop. (4 marks)

(b) Describe TWO actions to take when a learner sustains a minor cut in the workshop. (2 marks)

(c) State TWO reasons why tools should be stored properly after use. (2 marks)

13. Energy and simple electrical systems (7 marks)

(a) State THREE components that may be found in a simple electric circuit. (3 marks)

(b) Explain the difference between a conductor and an insulator, giving ONE example of each. (2 marks)

(c) State TWO precautions to observe when using electrical appliances. (2 marks)

14. Production, finishing and environmental care (7 marks)

(a) Give THREE reasons why a product may require surface finishing. (3 marks)

(b) Describe TWO ways of reducing waste during production. (2 marks)

(c) State TWO safe and environmentally responsible ways of handling workshop waste. (2 marks)

MARKING SCHEME – GRADE 9 PRE-TECHNICAL STUDIES CAT 2

Total: 50 marks

SECTION A: SHORT-ANSWER QUESTIONS (20 MARKS)

1. A specification is a clear statement of the requirements or features that a product must meet, such as size, function, material, safety and appearance. (2)

2. Any TWO: ensures the product is strong and durable; improves safety; controls cost; ensures materials suit the product's function; reduces waste; improves appearance. (1 each)

3. Any TWO: pencil/marker; scriber; marking gauge; try square; divider/compass; chalk line. (1 each)

4. Any TWO: strong; workable/easy to shape; attractive grain; available in different sizes; can be joined and finished; durable when suitably selected and treated. (1 each)

5. A first-aid box contains supplies used to give immediate care to an injured person; a fire extinguisher is used to control or put out a small fire of a suitable class. (1 each)

6. Any TWO correct examples: electric iron – heat; bulb – light; fan – movement/kinetic energy; radio/speaker – sound; electric kettle – heat. (1 each)

7. Any TWO: keep hands dry; avoid damaged cables; switch off and unplug before cleaning/repair; do not overload sockets; use correctly rated fuses; keep liquids away; report faults. (1 each)

8. The process of checking materials or products during and after production to ensure that they meet required standards or specifications. (2)

9. Any TWO: improves appearance; protects timber from moisture and wear; makes cleaning easier; increases durability; provides a smoother surface and reduces splinters. (1 each)

10. Any TWO: separate recyclable and non-recyclable waste; reuse suitable offcuts; dispose of hazardous waste safely; prevent chemicals entering drains/soil; follow local waste rules; reduce unnecessary waste. (1 each)

SECTION B: STRUCTURED QUESTIONS (30 MARKS)

11. Product design and material selection

(a) Any THREE, 1 mark each: strength/durability; cost; availability; ease of working; weight; appearance; safety; environmental impact; suitability for the intended function.

(b) Any THREE in a sensible sequence, 1 mark each: interpret drawing and dimensions; select and prepare materials; mark out; cut/shape components; join/assemble; finish surfaces; inspect/test the product.

(c) Any TWO: measure against drawing/specifications; test function and stability; inspect joints and finish; check safety and user requirements; compare product with design brief. (1 each)

12. Workshop safety and first aid

(a) Any FOUR: careless handling of tools; poor housekeeping; defective equipment; lack of PPE; horseplay; unsafe electrical connections; inadequate training; failure to follow instructions. (1 each)

(b) Any TWO: stop the activity and inform the teacher; wash hands/use gloves; apply gentle pressure with clean dressing to control bleeding; clean and cover a minor wound appropriately; seek adult/medical assistance if bleeding is severe or wound is deep. (1 each; do not reward unsafe treatment.)

(c) Any TWO: prevents accidents and injuries; prevents loss or damage; makes tools easy to find; maintains tools in good condition; keeps the work area orderly. (1 each)

13. Energy and simple electrical systems

(a) Any THREE: cell/battery or power source; connecting wires; switch; lamp/buzzer/motor/resistor/load. (1 each)

(b) A conductor allows electric current to pass through easily (e.g. copper); an insulator resists the flow of current (e.g. rubber or plastic). (1 each)

(c) Any TWO: do not touch appliances with wet hands; inspect plugs and cables; avoid overloading sockets; switch off after use; keep appliances away from water; use appliances according to instructions. (1 each)

14. Production, finishing and environmental care

(a) Any THREE: improve appearance; protect against moisture/corrosion/wear; increase durability; make surfaces smoother and safer; ease cleaning. (1 each)

(b) Any TWO: plan cutting layouts; measure accurately; use templates; reuse suitable offcuts; select correct stock sizes; avoid unnecessary reworking. (1 each)

(c) Any TWO: sort and recycle suitable materials; store and dispose of chemicals/solvents through approved procedures; keep sharp or hazardous waste in safe containers; do not burn or dump waste indiscriminately; follow local disposal guidance. (1 each)

Teacher’s note: This is original teacher-prepared assessment material. Confirm that the relevant Grade 9 Pre-Technical Studies content and safety procedures have been taught before administering. Accept other technically correct responses.